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Consumer-Lending Discrimination in the FinTech Era

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1 Big picture

- **Question.** Do Black and Latinx borrowers pay higher mortgage interest rates than risk-equivalent non-minority borrowers in the post-crisis mortgage market, and does algorithmic or platform lending reduce these disparities?
- **Setting.** The paper studies U.S. mortgage loans securitized by Fannie Mae and Freddie Mac, called GSE loans, and loans insured by the Federal Housing Administration, called FHA loans. These markets are useful because credit-risk pricing is largely determined by observable institutional rules rather than by lender-specific private information.
- **Legal idea.** U.S. fair-lending law allows lenders to use variables that are related to creditworthiness, even if their use has disparate effects. But courts have limited this “legitimate business necessity” defense to creditworthiness. Charging more because a borrower is less likely to shop around, lives in a financial-services desert, or can be profitably targeted is not a creditworthiness justification.
- **Core identification insight.** In GSE lending, credit-risk charges are determined by an 8×8 grid of credit-score and loan-to-value buckets. Once the researcher controls for that grid, plus time, lender, cash-out refinance status, and loan size, any remaining minority rate gap is not supposed to reflect credit risk transferred to the GSEs. FHA lending has little risk-based pricing, and the risk-pricing that exists is also tied mainly to LTV and credit score.
- **Data contribution.** The authors build a large loan-level dataset by merging HMDA, ATTOM, McDash, and Equifax data for 2009–2015. The core sample includes about **3.4 million GSE loans** and **2.3 million FHA loans**. They also use expanded **2018–2019 HMDA** data, which include points and total loan costs, for about **2.2 million GSE loans** and **1.0 million FHA loans**.
- **Main pricing result.** In the 2009–2015 core sample, Black and Latinx borrowers pay about **4.7–4.9 basis points more** on purchase mortgages and about **1.5–1.6 basis points more** on refinance mortgages, conditional on the paper’s controls.
- **Economic magnitude.** A few basis points can be economically meaningful in mortgage origination. The paper uses the heuristic that 20 basis points in interest rate is roughly one point, or 1% of the loan amount. Under that conversion, a 2 basis point rate gap corresponds to about 0.1% of the loan balance.
- **FinTech result.** FinTech lenders do not eliminate disparities. For 2009–2015 GSE loans, FinTech and non-FinTech rate gaps are similar. FinTech gaps are lower for FHA purchase and refinance loans. In 2018–2019, after controlling for total loan costs, FinTech disparities are again similar for most loan categories, except FHA refinance loans where the FinTech gap is not statistically different from zero.
- **Geography result.** Rate disparities are especially large in high-minority-share census tracts. Moreover, average rates are higher for all borrowers in such tracts, and minority borrowers in those tracts pay still more than non-minority borrowers. A minority borrower with a GSE purchase loan in a decile-10 minority-share tract pays about **13.8 basis points** more than a comparable non-minority borrower in a decile-1 tract; the analogous FHA purchase gap is about **16.2 basis points**.

- **Robustness.** The results survive controls and sample splits for put-back risk, realized default, servicing-cost proxies, minority-status measurement, lender-paid mortgage insurance, points, and total loan costs.
- **Takeaway.** Algorithmic lending can reduce direct face-to-face bias, but it does not necessarily remove discrimination. If pricing algorithms learn which borrowers or neighborhoods have less ability or tendency to shop, algorithmic pricing can reproduce or amplify disparities unrelated to creditworthiness.

2 Data and measurement (key objects)

2.1 Institutional setting: GSE and FHA mortgage pricing

The paper’s design is built around the way credit risk is priced and transferred in the GSE and FHA markets.

For **GSE loans**, a lender submits borrower and loan information to the GSE automated underwriting systems. If the loan is approved and originated, the lender can sell it to Fannie Mae or Freddie Mac. The GSE charges a guarantee fee, including loan-level price adjustments, to absorb credit risk. The key risk-pricing object is an 8×8 matrix:

$$\text{LLPA} = f(\text{credit score bucket, LTV bucket}). \quad (1)$$

The grid is not necessarily the actuarially best default model. That is not the point. The point is institutional: the grid determines what the lender must pay to transfer credit risk to the GSE. Therefore, if two borrowers are in the same grid cell and have comparable loan characteristics, a residual difference in the lender’s quoted rate is not explained by the GSE’s credit-risk price.

For **FHA loans**, the FHA insures loans against default. FHA loans have much less risk-based pricing than GSE loans. The risk pricing that exists is also mainly tied to LTV and/or credit score. The authors therefore apply a parallel grid-based control structure to FHA loans for comparability.

2.2 Core 2009–2015 merged sample

The core dataset merges four sources:

- **HMDA:** borrower race, ethnicity, income, loan purpose, loan amount, lender, and application information;
- **ATTOM:** property and loan-level records, including borrower names used to improve race and ethnicity classification when HMDA fields are missing;
- **McDash:** loan-level mortgage terms from Black Knight, including interest rates, LTVs, and monthly performance;
- **Equifax:** credit performance and borrower credit information pre-merged with McDash.

The sample focuses on first-lien, fixed-rate, owner-occupied, 30-year, single-family residential mortgages that are either GSE securitized or FHA insured. The paper excludes manufactured

housing, investment properties, condos, duplexes, triplexes, quadraplexes, and loans with outstanding second liens at origination.

Object	GSE loans	FHA loans
Number of loans	3,376,600	2,273,444
Mean interest rate	4.42%	4.40%
Mean LTV	74.0%	93.6%
Mean credit score	757.6	697.3
Minority borrower share	10.0%	24.6%
FinTech share	3.1%	1.6%
Refinance share	53.1%	26.2%

Interpretation. GSE borrowers have higher average credit scores and lower LTVs than FHA borrowers. FHA borrowers are more likely to be minority borrowers and tend to have higher leverage. This is why the design needs fine controls for credit-score/LTV buckets and separate estimates by GSE versus FHA and purchase versus refinance.

2.3 2018–2019 HMDA sample

The expanded 2018–2019 HMDA data include variables not available in the 2009–2015 HMDA data, especially:

- loan-level points paid, including negative points;
- total loan costs at origination;
- loan-level LTV;
- type of refinance;
- interest rate.

This second sample is important because a borrower can pay more points up front to get a lower interest rate, or receive lender credits in exchange for a higher rate. A rate gap alone could be misleading if minority borrowers simply choose different points-rate combinations. The 2018–2019 sample allows the paper to address this directly.

Object	GSE loans	FHA loans
Number of loans	2,212,246	953,192
Mean interest rate	4.44%	4.48%
Mean LTV	76.1%	89.7%
Mean proxy credit score	747.9	730.1
Minority borrower share	4.5%	14.8%
FinTech share	13.0%	9.1%
Refinance share	39.1%	30.3%
Mean points / loan balance	0.17%	0.17%
Mean total loan costs / balance	1.8%	3.7%

Measurement caveat. The 2018–2019 HMDA data do not provide the same merged individual credit-score information as the 2009–2015 merged dataset. The authors proxy for credit scores using average credit scores at the loan’s census tract and loan-type level, and they also control for income.

2.4 Minority borrower definition

The main minority indicator is:

$$\text{Minority}_i = \mathbf{1}\{\text{borrower } i \text{ is Black or Latinx}\}. \quad (2)$$

In the 2009–2015 data, the authors combine HMDA race/ethnicity variables with name-based race and ethnicity imputation from ATTOM borrower-name fields when HMDA values are missing. In robustness checks, they use only HMDA-observed race/ethnicity and also redefine treatment based on either the borrower or first co-borrower being Black or Latinx.

2.5 Outcome variable

The primary outcome is the mortgage interest rate in basis points:

$$\text{Rate}_i = 10000 \times \text{contract interest rate}_i. \quad (3)$$

The paper studies the interest rate on originated loans, conditional on obtaining credit. A separate appendix analysis considers rejection rates, but the authors emphasize that the rejection analysis is less well identified because rejected applicants do not have observed loan-level credit score and LTV in the same way originated borrowers do.

2.6 FinTech lender definition

The authors follow Buchak et al.’s platform-lender classification. The conceptual distinction is:

- **Traditional lenders:** rely more on loan officers, branches, or face-to-face interaction;
- **FinTech lenders:** rely more on platform processes, automation, and algorithmic scoring.

This distinction matters because FinTech could reduce discrimination by reducing human discretion, but it could also create algorithmic discrimination if predictive pricing models learn non-creditworthiness patterns correlated with race or ethnicity.

3 Models and empirical specifications (all equations + interpretation)

3.1 Mortgage-rate decomposition

The paper’s economic decomposition is:

$$\text{Rate}_i = \underbrace{\text{base market rate}}_{\text{time}} + \underbrace{\text{credit-risk price}}_{\text{GSE/FHA grid}} + \underbrace{\text{lender strategic markup}}_{\text{residual}}. \quad (4)$$

The first two components are controlled by time effects and the credit-score/LTV grid. The question is whether the residual strategic markup is systematically higher for Black and Latinx borrowers.

3.2 Baseline 2009–2015 pricing specification

The baseline regression is:

$$\begin{aligned} \text{Rate}_{it} = & \alpha \text{Minority}_i + \mu_{\text{Cash-out} \times \text{GSE grid} \times \text{year/month}} \\ & + \mu_{\text{Lender} \times \text{year/month}} + \mu_{\text{Amount decile}} + \varepsilon_{it}. \end{aligned} \quad (5)$$

Terms.

- Rate_{it} is the originated mortgage interest rate in basis points.
- Minority_i equals one if the borrower is Black or Latinx.
- $\text{Cash-out} \times \text{grid} \times \text{year/month}$ fixed effects absorb month-specific pricing for each credit-score/LTV grid cell and different treatment of cash-out refinances.
- $\text{Lender} \times \text{year/month}$ fixed effects absorb lender-specific rate sheets, market timing, and lender-level pricing strategies.
- $\text{Loan-amount-decile}$ fixed effects absorb the fact that smaller loans can have higher rates because origination has fixed costs.

Interpretation of α . The coefficient α is the minority rate differential among borrowers with similar institutional credit-risk prices, similar timing, similar lender-month pricing, similar loan size, and similar cash-out/grid status. Under the paper’s legal-identification argument, a positive α is evidence of discriminatory pricing because it is not justified by credit-risk pricing.

3.3 FinTech interaction specification

To compare FinTech and non-FinTech lenders, the paper estimates:

$$\begin{aligned} \text{Rate}_{it} = & \alpha_N (1 - \text{FinTech}_i) \text{Minority}_i + \alpha_F \text{FinTech}_i \text{Minority}_i \\ & + \mu_{\text{Cash-out} \times \text{grid} \times \text{year/month}} + \mu_{\text{Lender} \times \text{year/month}} + \mu_{\text{Amount decile}} + \varepsilon_{it}. \end{aligned} \quad (6)$$

Interpretation. α_N is the non-FinTech minority rate gap; α_F is the FinTech minority rate gap. Testing $\alpha_F = \alpha_N$ asks whether algorithmic/platform lenders materially reduce pricing disparities.

3.4 Geography with census-tract controls

The geography analysis adds census-tract-by-year or census-tract-by-year/month fixed effects:

$$\begin{aligned} \text{Rate}_{it} = & \alpha \text{Minority}_i + \mu_{\text{Cash-out} \times \text{grid} \times \text{year/month}} \\ & + \mu_{\text{Amount decile}} + \eta_{\text{Census tract} \times \text{year}} + \varepsilon_{it}. \end{aligned} \quad (7)$$

The paper also lets the minority coefficient vary by census-tract minority-share decile:

$$\text{Rate}_{it} = \sum_{d=1}^{10} \alpha_d \text{Minority}_i \times \mathbf{1}\{\text{tract minority-share decile} = d\} + \text{controls} + \varepsilon_{it}. \quad (8)$$

Interpretation. Census-tract fixed effects compare minority and non-minority borrowers within the same local area. The decile interaction asks whether discrimination is stronger in areas with larger minority shares.

3.5 Points and total loan costs in 2018–2019

Using the expanded HMDA data, the paper estimates specifications of the form:

$$\begin{aligned} \text{Rate}_i = & \alpha \text{Minority}_i + \mu_{\text{Lender} \times \text{year}} + \mu_{\text{Cost or points decile} \times \text{year}} \\ & + \mu_{\text{Cash-out} \times \text{LTV decile} \times \text{year}} + \mu_{\text{Cash-out} \times \text{credit decile} \times \text{year}} \\ & + \mu_{\text{Amount decile} \times \text{year}} + \mu_{\text{Cash-out} \times \text{income decile} \times \text{year}} + \varepsilon_i. \end{aligned} \quad (9)$$

The key variants are:

- no controls for points or costs;
- controls for points-paid decile by year;
- controls for total-loan-cost decile by year.

Interpretation. If minority borrowers paid higher rates only because they paid fewer points up front, then the minority coefficient should shrink or disappear after controlling for points or total costs. It does not.

3.6 Lender-specific mortgage-menu slopes

A concern raised in related work is that different lenders may offer different points-rate menus. Pooling lenders can create econometric problems if borrowers sort across menus. To address this, the paper allows each lender-year to have its own slope relating total loan cost to the interest rate:

$$\text{Rate}_i = \alpha \text{Minority}_i + \beta_{\ell(i), y(i)} \text{Cost}_i + \mu_{\ell(i) \times y(i)} + \text{other controls} + \varepsilon_i. \quad (10)$$

Interpretation. This specification compares minority and non-minority borrowers while allowing each lender-year to have its own mortgage menu level and cost-rate slope. The minority coefficient remains positive.

3.7 Rejection-rate analysis

The appendix also studies whether minority applicants are more likely to be rejected:

$$\Pr(\text{Reject}_i = 1) = F(\alpha \text{Minority}_i + \text{controls}). \quad (11)$$

The authors treat this evidence as suggestive because rejected loans lack some originated-loan variables, especially loan-level credit score and LTV. The pricing evidence is the central, more credible analysis.

4 Identification and credibility

4.1 What makes the design credible

The identification strategy is not a randomized experiment and not an instrument. Its credibility comes from a legal-institutional feature of mortgage markets:

GSE and FHA rules determine how credit risk is priced and transferred. Conditional on the relevant grid and controls, remaining differences in rates are not supposed to reflect the price of credit risk.

This is especially useful because a classic problem in lending-discrimination studies is omitted lender information. A lender might observe credit-relevant facts that the econometrician does not. Here, the authors argue that the relevant legitimate-business-necessity variables are captured because the GSE/FHA pricing framework determines the credit-risk charge.

4.2 What the estimate means legally

The paper's concept of discrimination is tied to disparate-impact fair-lending law. The claim is not necessarily that every loan officer intentionally discriminates. The claim is that, if minority borrowers pay more for reasons unrelated to creditworthiness, and if lenders cannot justify that pricing by legitimate business necessity, the resulting disparity is discriminatory under the legal framework used in the paper.

This distinction matters for FinTech. An algorithm can create a disparity without animus. For example, it may learn that some borrower profiles shop less, negotiate less, or live in less competitive local markets. If those profiles are correlated with race or ethnicity, and the pricing variable is not creditworthiness-related, the resulting disparity can still be legally problematic.

4.3 Put-back risk

A possible objection is that originators remain exposed to put-back risk if a loan later violates documentation or underwriting requirements. If minority borrowers generated more put-back risk, higher rates might reflect risk retained by the lender.

The paper addresses this in three ways:

1. focus on loans issued from 2013 to 2015, after put-back risk became less salient;

2. focus on high-quality borrowers for whom put-back risk should be especially low;
3. compare bank and nonbank lenders, because banks may have more franchise value and thus different exposure to put-back concerns.

The minority coefficients remain positive and similar in magnitude, so put-back risk is unlikely to explain the results.

4.4 Servicing-cost risk

Another objection is that servicers may bear costs when a borrower becomes delinquent, even if the GSE or FHA absorbs credit losses. If minority loans are more costly to service, higher rates might reflect expected servicing costs.

The paper directly controls for ex-post default outcomes, including foreclosure/REO, 60-days-plus delinquency, and 90-days-plus delinquency. The minority rate gap barely changes. The paper also studies heterogeneity by credit-score buckets, LTV buckets, and income deciles. The gap is higher for high-LTV borrowers and lower for higher-income borrowers, which may be partly consistent with servicing costs. But the weak relationship with realized default and credit score suggests another channel, such as shopping behavior or financial sophistication, may be more important.

4.5 Measurement of race and ethnicity

The authors combine HMDA self-reports with name-based imputation to increase sample coverage. That raises a measurement concern: imputed minority status could be wrong and correlated with pricing.

The paper addresses this by re-estimating the main regressions:

- using only observations with HMDA-provided race or ethnicity;
- defining treatment as either borrower or first co-borrower being Black or Latinx.

The results remain similar, indicating that the main findings are not driven by race/ethnicity classification errors.

4.6 Discount points and loan costs

Another important concern is that interest rates alone are not the full price of the mortgage. Borrowers may choose different combinations of points and interest rates.

The 2018–2019 HMDA data directly address this. Minority rate gaps remain positive and significant after controlling for points and total loan costs. The paper also uses specifications with lender-specific total-cost slopes, which address concerns about different lender points-rate menus.

4.7 Geography and local competition

The geography results are among the most economically informative parts of the paper. The paper finds two effects in high-minority-share census tracts:

1. rates are higher for all borrowers in high-minority-share tracts;
2. conditional on local area, minority borrowers pay more than non-minority borrowers.

This pattern is consistent with a channel based on local market power, financial-services deserts, low shopping intensity, or algorithmic pricing based on geography and behavior. It is less consistent with pure credit-risk pricing, because the design already controls for the relevant credit-risk grid.

4.8 Limitations

The main limitations are:

- the estimates apply to originated loans, so they do not fully identify discrimination in approval or rejection;
- the design is strongest for GSE loans, where the pricing grid is explicit;
- FHA pricing is less grid-like, although FHA risk pricing is also limited and largely tied to LTV and credit score;
- the paper identifies rate disparities and rules out many credit-risk explanations, but it does not definitively distinguish between negotiation, local competition, lender algorithms, branch pricing, and borrower shopping behavior.

5 Tables and figures

5.1 Figure 1: Example of the GSE grid

Read: Figure 1 shows Fannie Mae’s loan-level price adjustment grid. Fees vary by LTV and credit-score bucket. The grid is the institutional object that determines the GSE credit-risk charge.

Economic message: The paper’s design uses this grid as a control for credit-risk pricing. Conditional on grid cell, residual rate differences should not be credit-risk differences transferred to the GSE.

5.2 Table 1: Summary statistics for the 2009–2015 core sample

Read: The sample contains 3.38 million GSE loans and 2.27 million FHA loans. FHA borrowers have higher LTVs, lower credit scores, lower incomes, and a larger minority share than GSE borrowers.

Economic message: The two markets serve different borrower pools. The analysis therefore estimates results separately for GSE versus FHA and for purchase versus refinance loans.

5.3 Table 2: Summary statistics for the 2018–2019 HMDA sample

Read: The expanded HMDA sample contains 2.21 million GSE loans and 0.95 million FHA loans. It includes points and total loan costs, allowing the authors to test whether rate gaps are offset by different up-front costs.

Table 2: All Eligible Mortgages (Excluding MCM): LPA by Credit Score/LTV									
PRODUCT FEATURE	LPA by LTV Range								SFC
	≤ 60.00%	60.01 – 70.00%	70.01 – 75.00%	75.01 – 80.00%	80.01 – 85.00%	85.01 – 90.00%	90.01 – 95.00%	95.01 – 97.00%	
Representative Credit Score	Applicable for all mortgages with greater than 15 year terms For whole loans purchased on or before March 31, 2011, or loans delivered into MDS pools with issue dates of March 1, 2011 or earlier								
≥ 740	-0.250%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	0.000%	N/A
720 – 730	-0.250%	0.000%	0.000%	0.250%	0.000%	0.000%	0.000%	0.000%	N/A
700 – 710	-0.250%	0.500%	0.500%	0.750%	0.500%	0.500%	0.500%	0.500%	N/A
680 – 690	0.000%	0.500%	1.000%	1.500%	1.000%	0.750%	0.750%	0.500%	N/A
660 – 670	0.000%	1.000%	2.000%	2.500%	2.250%	1.750%	1.750%	1.250%	N/A
640 – 650	0.500%	1.250%	2.500%	3.000%	2.750%	2.250%	2.250%	1.750%	N/A
620 – 630	0.500%	1.500%	3.000%	3.000%	3.000%	2.750%	2.750%	2.500%	N/A
< 620 ¹⁰	0.500%	1.500%	3.000%	3.000%	3.000%	3.000%	3.000%	3.000%	N/A

Fig. 1. Example of the GSE grid. Presented is the LPA (loan-level price adjustment) grid of Fannie Mae for 2011. The figure is from the Fannie Mae Selling Guide, dated December 23, 2010. (MCM, now retired, refers to “My Community Mortgages,” a program of subsidized loans for low-income target areas.) The LPA grid has a parallel grid at Freddie Mac called the Credit Fees in Price chart. These grids provide the additional g-fee (guarantee fee) that lenders must pay the GSE for guaranteeing the mortgage, varying by LTV and credit score.

Table 1

Summary statistics: core sample. Data are fixed-rate mortgage originations obtained from a loan-level merge of HMDA, ATTOM, McDash, and Equifax data. Loan amount, applicant income and Latinx/Black are from HMDA. Interest rate, LTV, and credit score are from McDash-Equifax. Top-25-volume lender is calculated annually from volume of loans by lender. FinTech is a platform-lender identifier from Buchak et al. (2018).

	count	mean	sd	min	max
(a)GSE loans					
Cash-out refinance	3,376,600	0.0822052	0.2746771	0	1
CRA census tract	3,375,949	0.0926474	0.2899378	0	1
Credit score	2,950,931	757.6442	43.06677	620	850
FinTech	3,376,600	0.0312394	0.1739641	0	1
Income	3,252,686	101.9811	81.34873	20	9755
Loan amount	3,376,600	239.9792	121.9767	40	729
Loan interest rate	3,376,600	0.0442447	0.0054665	0.0275	0.07875
Loan-to-value ratio	3,376,600	0.7397208	0.1488221	0.3	0.95
Minority borrower	3,376,600	0.1001579	0.3002104	0	1
Refinance	3,376,600	0.5309895	0.4990388	0	1
Top-25 lender	3,376,600	0.4095857	0.4917574	0	1
N	3,376,600				
	count	mean	sd	min	max
(b)FHA loans					
Cash-out refinance	2,273,444	0.0212822	0.1443237	0	1
CRA census tract	2,273,365	0.1781905	0.3826731	0	1
Credit score	1,994,340	697.3241	49.53322	580	850
FinTech	2,273,444	0.0160492	0.1256648	0	1
Income	1,999,753	66.4546	45.83809	20	7424
Loan amount	2,273,444	176.7100	91.61529	40	729
Loan interest rate	2,273,444	0.0440192	0.0068095	0.0275	0.0775
Loan-to-value ratio	2,273,444	0.9358196	0.0674404	0.3	0.9825
Minority borrower	2,273,444	0.2461666	0.4307768	0	1
Refinance	2,273,444	0.2619699	0.4397065	0	1
Top-25 lender	2,273,444	0.2854264	0.4516174	0	1
N	2,273,444				

Economic message: The 2018–2019 sample is the paper’s answer to the most important price-measurement critique: maybe minority borrowers pay higher rates because they choose lower points. The later data show that this does not eliminate the gap.

Table 2

Summary statistics: 2018/19 HMDA data. Data are fixed-rate mortgage originations obtained from the 2018–2019 HMDA data. The credit scores are measured as average credit scores at the loan’s census tract for originated mortgages in the ATTOM/McDASH data. The borrower minority status is constructed from the HMDA measure for ethnicity and race in the 2018–2019 HMDA data.

	count	mean	sd	min	max
(a) GSE loans					
Credit score	2,212,246	747.86608	14.674482	620	829
FinTech	2,212,246	0.13038062	0.33672179	0	1
Income	2,184,873	100.70703	63.440554	20	635
Loan amount	2,212,246	263.61896	129.73451	45	725
Loan interest rate	2,212,246	0.04441362	0.00538968	0.0275	0.079
Loan-to-value ratio	2,212,246	0.76053186	0.14402249	0.3003096	0.94957983
Minority borrower	2,212,246	0.04495567	0.20720685	0	1
Points	2,164,245	0.00169405	0.0066614	−0.01922169	0.02753803
Refinance	2,185,338	0.39142412	0.48806904	0	1
Total loan costs	2,192,384	0.01818049	0.01120778	0	0.07018133
N	2,212,246				
	count	mean	sd	min	max
(b)FHA loans					
Credit score	953,192	730.0895	18.10742	580,125	826
FinTech	953,192	0.0912523	0.2879678	0	1
Income	910,670	78.64494	41.06824	20	635
Loan amount	953,192	232.0056	105.3531	45	725
Loan interest rate	953,192	0.0448173	0.006458	0.0275	0.07875
Loan-to-value ratio	953,192	0.8967726	0.0928592	0.300813	0.9823009
Minority borrower	953,192	0.1483636	0.3554602	0	1
Points	920,361	0.0016903	0.0072352	−0.0192209	0.0275377
Refinance	943,124	0.3030906	0.4595943	0	1
Total loan costs	901,869	0.0366662	0.0117278	0	0.0701836
N	953,192				

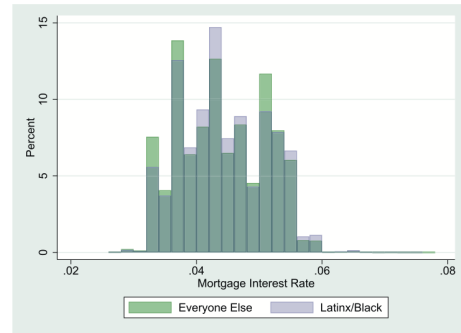
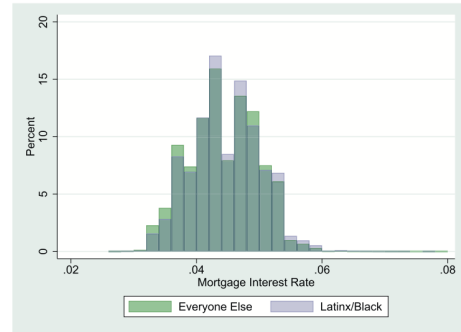


Fig. 2. Raw interest rates by race/ethnicity. Presented are histograms of raw interest rates originated on 30-year fixed-rate mortgages, 2009–2015. The histograms are plotted for Latinx/Black and for everyone else.

5.4 Figures 2 and 3: Raw and grid-demeaned interest-rate histograms

Read: Figure 2 shows wide raw interest-rate dispersion by race/ethnicity. Figure 3 subtracts the relevant GSE-grid/month-year mean and shows much tighter distributions, but still with residual variation.

Economic message: The grid removes a large amount of credit-risk and time variation. The remaining question is whether residual pricing differs systematically by minority status.

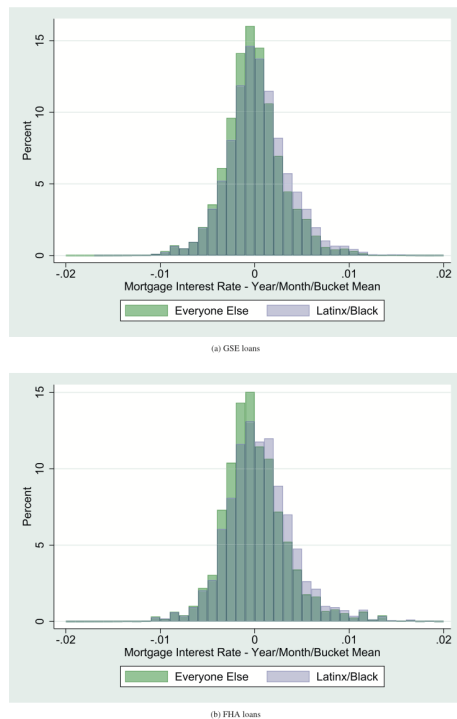


Fig. 3. De-meaned interest rate histograms by race/ethnicity. The role of the GSE grid. The figure shows loan interest rates for 30-year mortgages from 2009 to 2015, de-meaned to the GSE grid for the relevant month and year (calculated separately for purchase and refinance loans and for GSE vs. FHA loans). The histogram is plotted for Latinx/Black borrowers and for everyone else.

Table 3
Interest-rate differentials. The dependent variable is the interest rate on originated fixed-rate mortgages in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise. Fixed effects are included for cash-out $\times$ GSE-grid bucket $\times$ year/month, lender $\times$ year/month, and loan-amount decile. Standard errors are clustered at the lender level. ***, **, and * indicate significance at the 1%, 5%, and 10% conventional levels.

VARIABLES	GSE Loans		FHA Loans	
	Purchase (1) Interest rate	Refinance (2) Interest rate	Purchase (3) Interest rate	Refinance (4) Interest rate
Minority borrower	4.674*** (0.255)	1.632*** (0.227)	4.866*** (0.333)	1.527*** (0.253)
Observations	1,371,629	1,540,939	1,533,532	436,420
R-squared	0.803	0.769	0.854	0.869
Lender $\times$ year/month FE	Y	Y	Y	Y
Cash-out $\times$ bucket $\times$ year/month FE	Y	Y	Y	Y
Amount decile FE	Y	Y	Y	Y

Robust standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.5 Table 3: Baseline interest-rate differentials

Read: The minority coefficient is positive and significant in all four columns:

Loan category	Minority rate gap	Interpretation
GSE purchase	4.674 bps	Large purchase-market gap
GSE refinance	1.632 bps	Smaller but significant
FHA purchase	4.866 bps	Similar to GSE purchase
FHA refinance	1.527 bps	Smaller but significant

Economic message: Minority borrowers pay more even after controlling for credit-risk grid, lender-month effects, loan size, and cash-out/grid timing. The purchase-loan gap is much larger than the refinance gap, consistent with weaker borrower experience or greater urgency in purchase markets.

5.6 Table 4: FinTech versus non-FinTech lenders, 2009–2015

Read: For GSE loans, FinTech and non-FinTech minority gaps are similar. For FHA loans, FinTech gaps are smaller: about 3.55 bps for FHA purchase and 0.97 bps for FHA refinance, compared with 4.88 and 1.55 bps for non-FinTech.

Economic message: FinTech does not automatically remove disparities. It may reduce them in some FHA segments, but GSE disparities remain comparable across FinTech and non-FinTech lenders.

Table 4
Interest-rate differentials: FinTech vs. non-FinTech lenders. The table reports interest-rate differentials as in Table 3, split by FinTech vs. non-FinTech lenders (as defined by Buchak et al., 2018). The dependent variable is the interest rate on originated fixed-rate mortgages in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise. Fixed effects are included for cash-out $\times$ GSE-grid bucket $\times$ year/month, lender $\times$ year/month, and loan-amount decile. Standard errors are clustered at the lender level. ***, **, and * indicate significance at the 1%, 5%, and 10% conventional levels.

VARIABLES	GSE Loans		FHA Loans	
	Purchase (1) Interest rate	Refinance (2) Interest rate	Purchase (3) Interest rate	Refinance (4) Interest rate
Non-FinTech $\times$ Minority	4.666*** (0.256)	1.631*** (0.238)	4.877*** (0.336)	1.548*** (0.262)
FinTech $\times$ Minority	5.081*** (0.124)	1.565*** (0.271)	3.550*** (0.373)	0.969** (0.385)
Observations	1,371,629	1,540,939	1,533,532	436,420
R-squared	0.803	0.769	0.854	0.869
p-value for test of equality	0.1172	0.8570	0.0084	0.2204
Lender $\times$ year/month FE	Y	Y	Y	Y
Cash-out $\times$ bucket $\times$ year/month FE	Y	Y	Y	Y
Amount decile FE	Y	Y	Y	Y
FinTech FE	Y	Y	Y	Y

Robust standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

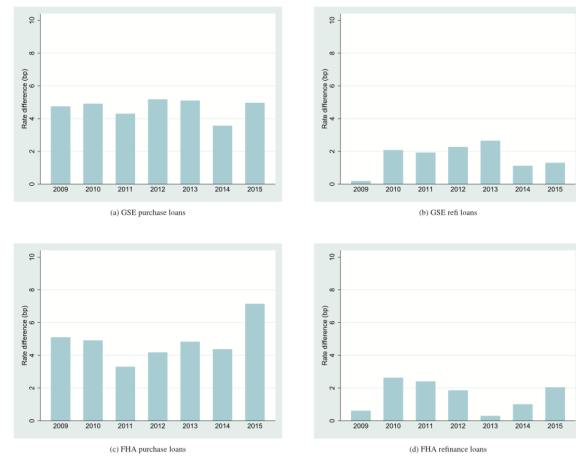


Fig. 4. Interest-rate differentials by year. The figure plots interest-rate differentials as in Table 3, estimated year by year. The dependent variable is the interest rate on originated fixed-rate mortgages in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise. Fixed effects are included for cash-out $\times$ GSE-grid bucket $\times$ year/month, lender $\times$ year/month, and loan-amount decile.

5.7 Figure 4: Interest-rate differentials by year

Read: The estimated minority gaps are fairly stable from 2009 to 2015.

Economic message: The effects are not driven by one unusual crisis-year or by an obvious trend around the rise of FinTech lending or changes in loan-originator compensation rules.

5.8 Table 5 and Figure 5: Census-tract controls and minority-share deciles

Read: Adding census-tract-by-year controls reduces the baseline coefficients but does not remove them. In Table 5, the gaps are about 2.00 bps for GSE purchase, 0.77 bps for GSE refinance, 1.93 bps for FHA purchase, and 0.50 bps for FHA refinance. Figure 5 shows the gap tends to be larger in high-minority-share tracts.

Economic message: Part of the raw disparity is geographic: minority borrowers are more likely to live in higher-rate areas. But even within local areas, minority borrowers pay more.

5.9 Figure 6: Average rate levels by minority-share decile

Read: Average mortgage rates increase sharply with census-tract minority share. Rates in decile-10 minority-share tracts exceed rates in decile-1 tracts by 9.7–14.3 bps for purchase loans and 3.1–5.8 bps for refinance loans.

Table 5
Interest-rate differentials with census-tract controls. The dependent variable is the interest rate on originated fixed-rate mortgages in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise. Fixed effects are included for cash-out $\times$ GSE-grid bucket $\times$ year/month, loan-amount decile, and census-tract $\times$ year/month. Standard errors are clustered at the lender level. ***, **, and * indicate significance at the 1%, 5%, and 10% conventional levels.

VARIABLES	GSE Loans		FHA Loans	
	Purchase (1)	Refinance (2)	Purchase (3)	Refinance (4)
Minority borrower	2.003*** (0.186)	0.768*** (0.207)	1.926*** (0.213)	0.499*** (0.183)
Observations	1,320,534	1,489,748	1,476,865	367,377
R-squared	0.827	0.788	0.870	0.894
Cash-out $\times$ bucket $\times$ year/month FE	Y	Y	Y	Y
Amount decile FE	Y	Y	Y	Y
Census tract $\times$ year FE	Y	Y	Y	Y

Robust standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

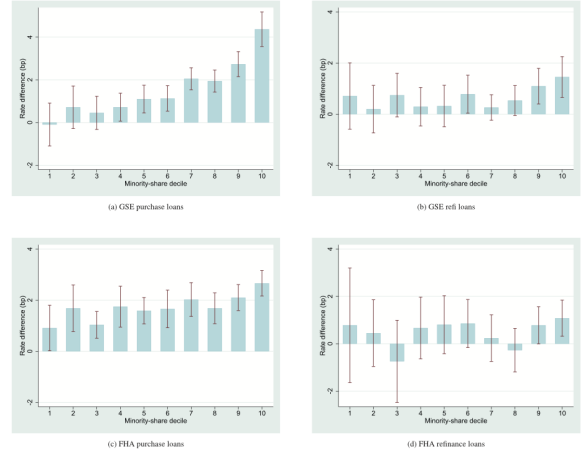


Fig. 5. Interest-rate differentials with census-tract controls by minority-share decile (point estimates and 95% confidence intervals). The dependent variable is the interest rate on originated fixed-rate mortgages in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise, interacted with indicator variables for minority-share decile. Fixed effects are included for cash-out $\times$ GSE-grid bucket $\times$ year/month, loan-amount decile, and census-tract $\times$ year. Standard errors are clustered at the lender level.

Economic message: Minority borrowers in high-minority areas face a double burden: higher local average rates plus an additional minority-specific gap within the area.

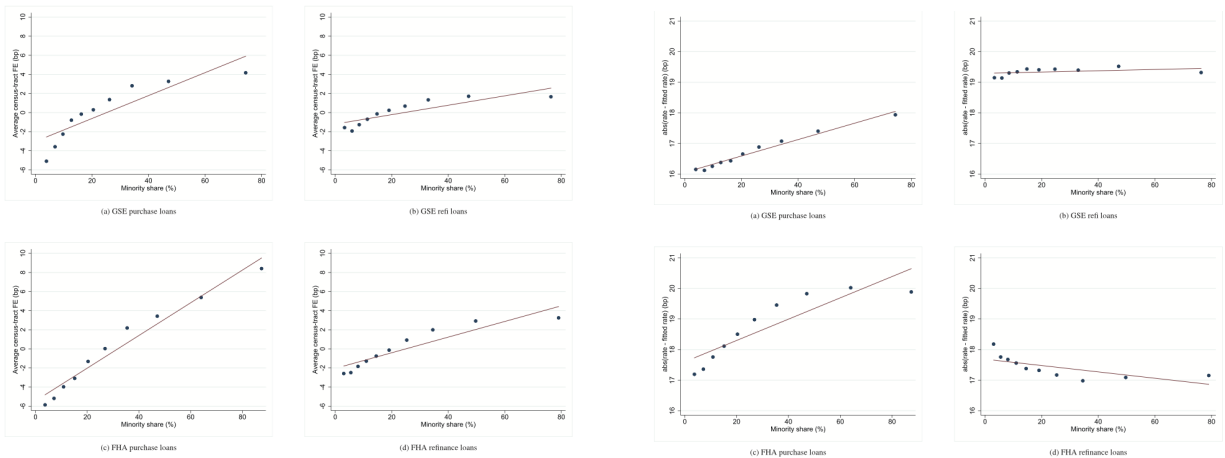


Fig. 6. Average interest-rate levels by minority-share decile. The figure shows binned scatter plots of the census-tract $\times$ year fixed effects from the regression in Table 5 versus the census-tract minority share, with the data in 10 equal bins.

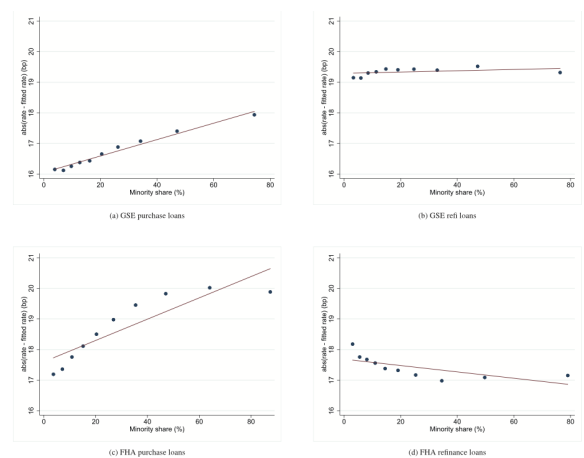


Fig. 7. Binned scatter plots of absolute residuals from Table 5 against minority share.

5.10 Figure 7: Absolute residuals and minority-share decile

Read: The dispersion of residual rates is higher in high-minority-share tracts, especially for purchase loans.

Economic message: A pure cost story would predict higher mean rates in costly areas, but not necessarily higher residual dispersion. The dispersion pattern points toward strategic pricing, competition, or shopping differences.

5.11 Figure 8: Lender minority-share deciles

Read: The minority coefficient generally rises with the lender's share of loans to minority borrowers, particularly for purchase loans.

Economic message: Disparities are larger in parts of the market where minority borrowers are

more concentrated. This is consistent with market segmentation or lender strategies targeted to particular borrower pools.

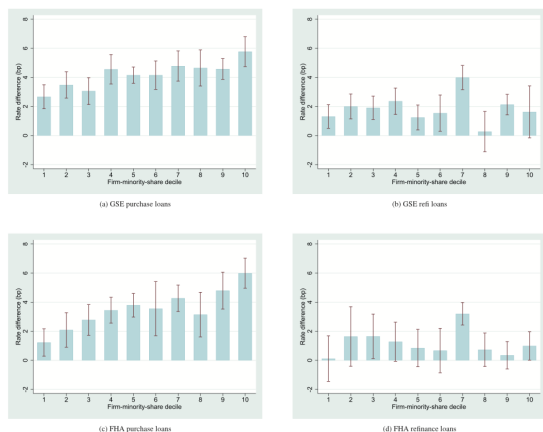


Fig. 8. Interest-rate differentials by firm-minority-share decile (point estimates and 95% confidence intervals). The dependent variable is the interest rate on originated fixed-rate mortgages in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise, interacted with indicator variables for firm-minority-share decile. Fixed effects are included for cash-out $\times$ GSE-grid bucket $\times$ year/month, lender $\times$ year/month, loan-amount decile, and firm-minority-share decile. Standard errors are clustered at the lender level.

Table 6
Interest-rate differentials by CRA status. The dependent variable is the interest rate on originated fixed-rate mortgages in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise, interacted with indicator variables for CRA status. Fixed effects are included for cash-out $\times$ GSE-grid bucket $\times$ year/month, lender $\times$ year/month, and loan-amount decile. Standard errors are clustered at the lender level. ***, **, and * indicate significance at the 1%, 5%, and 10% conventional levels.

VARIABLES	GSE Loans		FHA Loans	
	Purchase (1) Interest rate	Refinance (2) Interest rate	Purchase (3) Interest rate	Refinance (4) Interest rate
Non-CRA tract $\times$ Minority	4.043*** (0.248)	1.411*** (0.194)	4.555*** (0.338)	1.334*** (0.225)
CRA tract $\times$ Minority	6.431*** (0.334)	1.856*** (0.474)	5.397*** (0.328)	1.223*** (0.365)
Observations	1,371,437	1,540,614	1,533,484	436,401
R-squared	0.803	0.769	0.854	0.869
p-value for test of equality	0.0000	0.2188	0.0000	0.6579
Lender $\times$ year/month FE	Y	Y	Y	Y
Cash-out $\times$ bucket $\times$ year/month FE	Y	Y	Y	Y
Amount decile FE	Y	Y	Y	Y
CRA FE	Y	Y	Y	Y

Robust standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.12 Table 6: CRA versus non-CRA census tracts

Read: Minority gaps are positive in both CRA and non-CRA tracts. In CRA tracts, minority purchase borrowers pay 6.43 bps more on GSE loans and 5.40 bps more on FHA loans.

Economic message: The disparity is not confined to non-CRA areas. It persists in low- and moderate-income neighborhoods covered by community reinvestment concerns.

5.13 Table 7: Put-back risk robustness

Read: The minority coefficients remain positive in post-2012 loans, high-quality borrower samples, and bank versus nonbank comparisons. For example, post-2012 purchase gaps are 4.54 bps for GSE and 5.58 bps for FHA.

Economic message: The baseline gap is unlikely to be explained by originator exposure to put-back risk.

5.14 Table 8: Ex-post default controls

Read: Controlling for later foreclosure/REO, 60-days-plus delinquency, and 90-days-plus delinquency barely changes the minority coefficient. For example, the GSE purchase coefficient moves from 4.674 bps in Table 3 to 4.619 bps.

Economic message: Realized default-related outcomes do not explain the rate gaps. This weakens the servicing-cost and hidden-risk explanations.

Table 7

Interest-rate differentials. The dependent variable is the interest rate on originated fixed-rate mortgages in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise. Fixed effects are included for cash-out $\times$ GSE-grid bucket $\times$ year/month, lender $\times$ year/month, and loan-amount decile. Standard errors are clustered at the lender level. ***, **, and * indicate significance at the 1%, 5%, and 10% conventional levels.

VARIABLES	GSE Loans		FHA Loans	
	Purchase (1) Interest rate	Refinance (2) Interest rate	Purchase (3) Interest rate	Refinance (4) Interest rate
Minority borrower	4.539*** (0.260)	1.865*** (0.274)	5.583*** (0.470)	1.167*** (0.369)
Observations	690,659	374,700	544,112	111,098
R-squared	0.675	0.683	0.577	0.629
Lender $\times$ year/month FE	Y	Y	Y	Y
Cash-out $\times$ bucket $\times$ year/month FE	Y	Y	Y	Y
Amount decile FE	Y	Y	Y	Y

Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

(a) **Post-2012:** Loans issued from 2013 to 2015

VARIABLES	GSE Loans		FHA Loans	
	Purchase (1) Interest rate	Refinance (2) Interest rate	Purchase (3) Interest rate	Refinance (4) Interest rate
Minority borrower	3.359*** (0.367)	1.481*** (0.299)	3.715*** (0.877)	1.110 (2.031)
Observations	77,432	314,734	6441	3233
R-squared	0.859	0.788	0.892	0.907
Lender $\times$ year/month FE	Y	Y	Y	Y
Cash-out $\times$ bucket $\times$ year/month FE	Y	Y	Y	Y
Amount decile FE	Y	Y	Y	Y

Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

(b) **High-quality borrowers:** Credit score ≥ 740 (700) and LTV ≤ 0.6 (0.75) for GSE (FHA) loans.

VARIABLES	GSE Loans		FHA Loans	
	Purchase (1) Interest rate	Refinance (2) Interest rate	Purchase (3) Interest rate	Refinance (4) Interest rate
Bank $\times$ Minority	4.274*** (0.192)	1.516*** (0.320)	3.378*** (0.483)	1.672*** (0.599)
Nonbank $\times$ Minority	4.954*** (0.315)	1.745*** (0.257)	5.341*** (0.306)	1.509*** (0.225)
Observations	1,278,029	1,466,461	1,418,917	401,325
R-squared	0.806	0.769	0.857	0.870
p-value for test of equality	0.0487	0.5762	0.0001	0.7952
Lender $\times$ year/month FE	Y	Y	Y	Y
Cash-out $\times$ bucket $\times$ year/month FE	Y	Y	Y	Y
Amount decile FE	Y	Y	Y	Y
Nonbank FE	Y	Y	Y	Y

Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

(c) **Banks vs. non-banks**

Table 8

Interest-rate differentials controlling for ex-post default status. The dependent variable is the interest rate on originated fixed-rate mortgages in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise; along with controls for whether the loan subsequently went into foreclosure/REO, 60-days-plus delinquent, or 90-days-plus delinquent. Fixed effects are included for cash-out $\times$ GSE-grid bucket $\times$ year/month, lender $\times$ year/month, and loan-amount decile. Standard errors are clustered at the lender level. ***, **, and * indicate significance at the 1%, 5%, and 10% conventional levels.

VARIABLES	GSE Loans		FHA Loans	
	Purchase (1) Interest rate	Refinance (2) Interest rate	Purchase (3) Interest rate	Refinance (4) Interest rate
Minority borrower	4.619*** (0.254)	1.614*** (0.228)	4.787*** (0.332)	1.510*** (0.253)
Foreclosure/REO	2.468*** (0.613)	1.887*** (0.688)	0.588*** (0.227)	1.491*** (0.310)
60+ days delinquent	5.235*** (0.333)	4.297*** (0.442)	1.894*** (0.167)	0.148 (0.320)
90+ days delinquent	1.258*** (0.463)	1.864*** (0.443)	1.223*** (0.171)	1.184*** (0.352)
Observations	1,371,629	1,540,939	1,533,532	436,420
R-squared	0.803	0.769	0.854	0.869
Lender $\times$ year/month FE	Y	Y	Y	Y
Cash-out $\times$ bucket $\times$ year/month FE	Y	Y	Y	Y
Amount decile FE	Y	Y	Y	Y

Robust standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

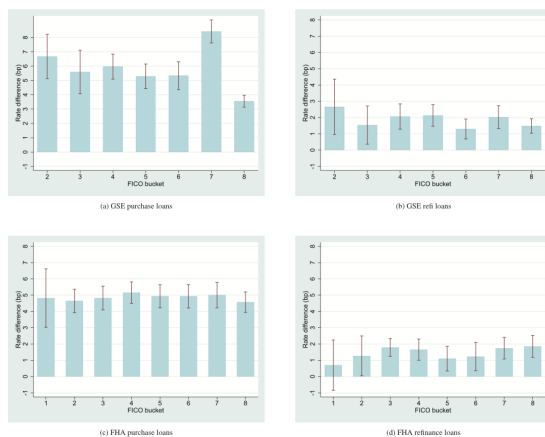


Fig. 9. Interest-rate differentials by GSE-credit-score bucket (point estimates and 95% confidence intervals). The dependent variable is the interest rate on originated fixed-rate mortgages in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise, interacted with indicator variables for credit-score bucket. Fixed effects are included for cash-out $\times$ GSE-grid bucket $\times$ year/month, lender $\times$ year/month, and loan-amount decile. Standard errors are clustered at the lender level.

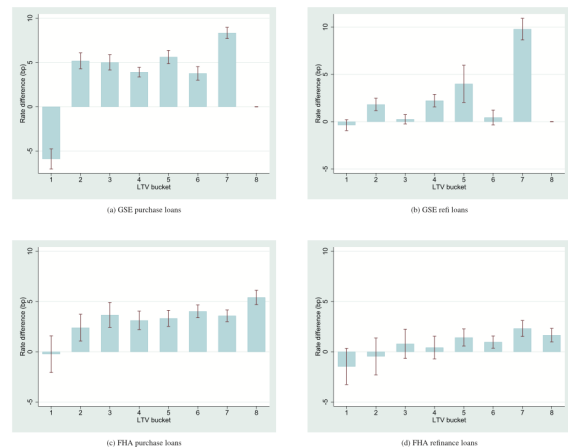


Fig. 10. Interest-rate differentials by GSE-LTV bucket (point estimates and 95% confidence intervals). The dependent variable is the interest rate on originated fixed-rate mortgages in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise, interacted with indicator variables for GSE-LTV bucket. Fixed effects are included for cash-out $\times$ GSE-grid bucket $\times$ year/month, lender $\times$ year/month, and loan-amount decile. Standard errors are clustered at the lender level.

5.15 Figures 9, 10, Table 9, and Figure 11: Credit score, LTV, mortgage insurance, and income

Read: The minority gap varies across credit-score, LTV, and income buckets. The gap is higher for high-LTV loans and tends to fall with borrower income. Table 9 restricts GSE loans to $LTV \leq 80\%$ and still finds significant gaps: 4.215 bps for purchase loans and 1.711 bps for refinance loans.

Economic message: Some heterogeneity is consistent with servicing costs or borrower sophistication, but the persistence of gaps across safer loans and lower-LTV loans shows the result is not driven by mortgage insurance or obvious risk buckets.

Table 9

Interest-rate differentials for GSE loans with $LTV \leq 80\%$. The dependent variable is the interest rate on originated fixed-rate mortgages in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise. Fixed effects are included for cash-out $\times$ GSE-grid bucket $\times$ year/month, lender $\times$ year/month, and loan-amount decile. Standard errors are clustered at the lender level. ***, **, and * indicate significance at the 1%, 5%, and 10% conventional levels.

VARIABLES	GSE Loans	
	Purchase (1) Interest rate	Refinance (2) Interest rate
Minority borrower	4.215*** (0.280)	1.711*** (0.219)
Observations	844,343	1,280,664
R-squared	0.835	0.778
Lender $\times$ year/month FE	Y	Y
Cash-out $\times$ bucket $\times$ year/month FE	Y	Y
Amount decile FE	Y	Y

Robust standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

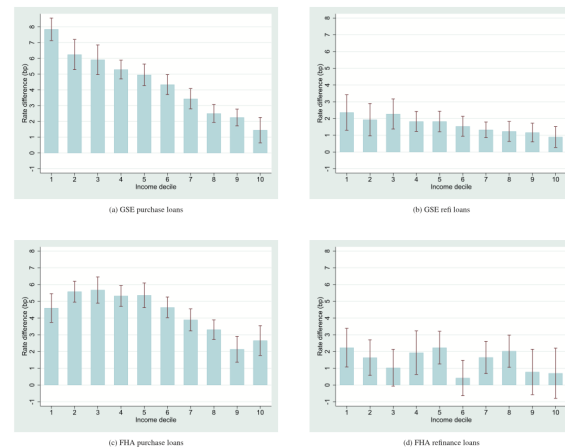


Fig. 11. Interest-rate differentials by income decile (point estimates and 95% confidence intervals). The dependent variable is the interest rate on originated fixed-rate mortgages in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise, interacted with indicator variables for income decile. Fixed effects are included for cash-out $\times$ GSE-grid bucket $\times$ year/month, lender $\times$ year/month, loan-amount decile, and income decile. Standard errors are clustered at the lender level.

5.16 Table 10: 2018–2019 HMDA, controlling for points and costs

Read: With controls for total loan costs, minority borrowers pay 7.709 bps more on GSE purchase loans, 6.806 bps more on GSE refinance loans, 5.441 bps more on FHA purchase loans, and 1.926 bps more on FHA refinance loans.

Economic message: The minority rate gap is not simply an artifact of points. Even after controlling for up-front loan costs, minority borrowers pay higher interest rates.

5.17 Table 11: FinTech lenders in 2018–2019

Read: For FinTech lenders, the minority gaps are 7.367 bps for GSE purchase, 7.357 bps for GSE refinance, 5.000 bps for FHA purchase, and -0.365 bps for FHA refinance. The FHA refinance estimate is not statistically significant.

Economic message: FinTech lending does not eliminate disparities in the main GSE and purchase-loan markets. The exception is FHA refinance lending, where the FinTech gap is not distinguishable from zero.

Table 10

Interest-rate differentials: 2018/2019 HMDA data controlling for points paid/total loan costs. The dependent variable is the interest rate on originated fixed-rate mortgages in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise. Fixed effects are included for lender $\times$ year, total-cost decile $\times$ year, cash-out $\times$ LTV decile $\times$ year, cash-out $\times$ census-tract credit-score decile $\times$ year, and loan-amount decile $\times$ year. Panel (a) does not control for either points paid or total loan costs, panel (b) controls for points paid, and panel (c) controls for total loan costs. Standard errors are clustered at the lender level. ***, **, and * indicate significance at the 1%, 5%, and 10% conventional levels.

VARIABLES	GSE Loans		FHA Loans	
	Purchase (1)	Refinance (2)	Purchase (3)	Refinance (4)
	Interest rate	Interest rate	Interest rate	Interest rate
Minority borrower	7.377*** (0.472)	5.998*** (0.392)	5.529*** (0.386)	1.632*** (0.541)
Observations	1,315,200	842,640	655,261	245,437
R-squared	0.376	0.483	0.341	0.356
Lender x year FE	Y	Y	Y	Y
Cash-out x LTV decile x year FE	Y	Y	Y	Y
Cash-out x credit decile x year FE	Y	Y	Y	Y
Amount decile x year FE	Y	Y	Y	Y
Cash-out x income decile x year FE	Y	Y	Y	Y

Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

(a) **No controls for points or costs.**

VARIABLES	GSE Loans		FHA Loans	
	Purchase (1)	Refinance (2)	Purchase (3)	Refinance (4)
	Interest rate	Interest rate	Interest rate	Interest rate
Minority borrower	7.820*** (0.482)	6.900*** (0.375)	5.530*** (0.384)	2.172*** (0.415)
Observations	1,315,200	842,640	655,261	245,437
R-squared	0.386	0.493	0.343	0.366
Lender x year FE	Y	Y	Y	Y
Point decile x year FE	Y	Y	Y	Y
Cash-out x LTV decile x year FE	Y	Y	Y	Y
Cash-out x credit decile x year FE	Y	Y	Y	Y
Amount decile x year FE	Y	Y	Y	Y
Cash-out x income decile x year FE	Y	Y	Y	Y

Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

(b) **Controlling for points paid.**

VARIABLES	GSE Loans		FHA Loans	
	Purchase (1)	Refinance (2)	Purchase (3)	Refinance (4)
	Interest rate	Interest rate	Interest rate	Interest rate
Minority borrower	7.709*** (0.520)	6.806*** (0.354)	5.441*** (0.394)	1.926*** (0.420)
Observations	1,306,553	835,769	631,631	240,015
R-squared	0.381	0.490	0.339	0.361
Lender x year FE	Y	Y	Y	Y
Cost decile x year FE	Y	Y	Y	Y
Cash-out x LTV decile x year FE	Y	Y	Y	Y
Cash-out x credit decile x year FE	Y	Y	Y	Y
Amount decile x year FE	Y	Y	Y	Y
Cash-out x income decile x year FE	Y	Y	Y	Y

Robust standard errors in parentheses

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(c) **Controlling for total loan costs.**

Table 11

Interest-rate differentials by FinTech firms: 2018/2019 HMDA data controlling for total loan costs.

The dependent variable is the interest rate on originated fixed-rate mortgages in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise. Fixed effects are included for lender $\times$ year, total-cost decile $\times$ year, cash-out $\times$ LTV decile $\times$ year, cash-out $\times$ census-tract credit-score decile $\times$ year, and loan-amount decile $\times$ year. Standard errors are clustered at the lender level. ***, **, and * indicate significance at the 1%, 5%, and 10% conventional levels.

VARIABLES	GSE Loans		FHA Loans	
	Purchase (1)	Refinance (2)	Purchase (3)	Refinance (4)
	Interest rate	Interest rate	Interest rate	Interest rate
Non-FinTech $\times$ Minority	7.751*** (0.595)	6.619*** (0.461)	5.466*** (0.414)	2.480*** (0.483)
FinTech $\times$ Minority	7.367*** (0.547)	7.357*** (0.222)	5.000*** (0.718)	-0.365 (0.244)
Observations	1,306,553	835,769	631,631	240,015
R-squared	0.381	0.490	0.339	0.361
Lender x year FE	Y	Y	Y	Y
Cost decile x year FE	Y	Y	Y	Y
Cash-out x LTV decile x year FE	Y	Y	Y	Y
Cash-out x credit decile x year FE	Y	Y	Y	Y
Amount decile x year FE	Y	Y	Y	Y
Cash-out x income decile x year FE	Y	Y	Y	Y
FinTech FE	Y	Y	Y	Y

Robust standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 12

Interest-rate differentials with firm-specific regression slope: 2018/2019 HMDA data controlling for total loan cost.

The dependent variable is the loan interest rate in basis points. The independent variable equals 1 if the borrower is Black or Latinx, and 0 otherwise. Also on the right is the total loan cost, trimmed at the 1% and 99% levels, with a separate slope for each lender $\times$ year combination. Fixed effects are included for lender $\times$ year, cash-out $\times$ LTV decile $\times$ year, cash-out $\times$ census-tract credit decile $\times$ year, loan-amount decile $\times$ year, and income $\times$ year. Standard errors are clustered at the lender level. ***, **, and * indicate significance at the 1%, 5%, and 10% conventional levels.

VARIABLES	GSE Loans		FHA Loans	
	Purchase (1)	Refinance (2)	Purchase (3)	Refinance (4)
	Interest rate	Interest rate	Interest rate	Interest rate
Minority borrower	7.683*** (0.534)	6.781*** (0.338)	5.520*** (0.390)	1.766*** (0.446)
Observations	1,303,784	834,678	620,341	231,615
R-squared	0.386	0.496	0.350	0.386
Lender x year FE	Y	Y	Y	Y
Lender x year cost slope	Y	Y	Y	Y
Cash-out x LTV decile x year FE	Y	Y	Y	Y
Cash-out x credit decile x year FE	Y	Y	Y	Y
Amount decile x year FE	Y	Y	Y	Y
Cash-out x income decile x year FE	Y	Y	Y	Y

Robust standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.18 Table 12: Lender-specific cost slopes

Read: Allowing each lender-year to have its own total-cost slope still yields positive and significant gaps: 7.683 bps for GSE purchase, 6.781 bps for GSE refinance, 5.520 bps for FHA purchase, and 1.766 bps for FHA refinance.

Economic message: The results are robust to heterogeneous lender mortgage menus. This directly addresses the concern that different points-rate schedules could create spurious pooled estimates.

5.19 Section 7 and Internet Appendix: Rejection-rate discrimination

Read: The paper finds suggestive evidence that minority applicants are more likely to be rejected than non-minority applicants, including among FinTech lenders.

Economic message: Pricing discrimination is not the only possible margin. However, because rejected applicants lack the same loan-level credit variables, the rejection evidence is weaker than the pricing evidence.

6 Short summary

This paper studies mortgage discrimination in the FinTech era by exploiting a special institutional feature of U.S. mortgage markets. In GSE lending, the price of credit risk is determined by a credit-score-by-LTV grid; in FHA lending, risk pricing is limited and mostly tied to the same variables. This makes it possible to compare minority and non-minority borrowers whose credit-risk prices should be equivalent.

The main result is that Black and Latinx borrowers pay higher mortgage rates. In the 2009–2015 merged sample, the gaps are about 4.7–4.9 basis points for purchase loans and 1.5–1.6 basis points for refinance loans. In the 2018–2019 HMDA sample, after controlling for points and total loan costs, the gaps are larger for most categories, especially GSE loans.

FinTech lending does not eliminate the problem. FinTech and non-FinTech gaps are similar for GSE mortgages, while FinTech gaps are lower for FHA refinance loans. The geography evidence is especially important: rates are higher in high-minority-share census tracts for all borrowers, and minority borrowers in those tracts pay an additional premium.

The evidence is robust to many alternative explanations, including put-back risk, ex-post default, servicing-cost proxies, measurement of minority status, lender-paid mortgage insurance, points, total loan costs, and heterogeneous lender cost-rate menus. The most plausible interpretation is that lenders, whether human or algorithmic, extract higher markups from borrowers or locations with lower shopping intensity or weaker competitive conditions. Because that explanation is unrelated to creditworthiness, it is problematic under the fair-lending framework used in the paper.

7 Conclusion

7.1 Core conclusion

Black and Latinx borrowers pay higher mortgage rates than otherwise comparable non-minority borrowers in GSE and FHA mortgage markets. The disparity is not fully explained by credit risk, points, loan costs, default, put-back risk, or measurement error.

7.2 Economic intuition

The paper's central intuition is that discrimination can arise through pricing discretion, not only through explicit denial of credit. Once credit risk is transferred through the GSE or FHA system, lenders still choose rate markups. Those markups may depend on borrower shopping behavior, negotiation, local competition, geography, or algorithmic predictions of willingness to pay. If these variables are correlated with race or ethnicity and are unrelated to creditworthiness, they can generate discriminatory pricing even without face-to-face animus.

7.3 Why the FinTech angle matters

The paper is not anti-algorithm. Rather, it shows that algorithmic lending is not automatically fair. Algorithms can reduce human bias, but they can also learn profitable non-credit pricing strategies. If the algorithm learns who is less likely to shop, who has fewer local alternatives, or which neighborhoods have weaker competition, it can produce disparities that look very similar to traditional discrimination.

7.4 Policy takeaway

The policy problem is not only whether lenders use race directly. It is whether pricing systems use variables or strategies that create minority disparities without a creditworthiness justification. Monitoring fair lending in the FinTech era therefore requires attention to algorithmic pricing, local market competition, borrower shopping frictions, points-rate menus, and the distinction between credit-risk prediction and profit-maximizing price discrimination.

7.5 Takeaway

Even in highly standardized, government-backed mortgage markets, Black and Latinx borrowers pay more for credit; FinTech reduces some face-to-face frictions but does not eliminate discriminatory pricing.